ISA AND CORE2350B (RP2350B)

Bus_ISA_8bit

PWR_FLAG

GND RESET +5V IRQ2 PWR_FLAG -5V DRQ2 -12V UNUSED +12V GND SMEMW SMEMR IOW IOR DACK3 DRQ3 DACK1 DRQ1 DACK0 CLK IRQ7 IRQ6 IRQ5 IRQ4 IRQ3 DACK2 TC ALE +5V VCC OSC GND

IO DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0 IO_READY AEN BA19 BA18 BA17 BA16 BA15 BA14 BA13 BA12 BA11 BA10 BA09 BA08 BA07 BA06 BA05 BA04 BA03 BA02 BA01 BA00

D7 D6 D5 D4 D3 D2 D1 D0 I/OCHRDY AEN A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

U12A

Core2350B

GPIO029 grounded as on v2.0 (board ID: the RP2040 build reads it with the ADC)

GPIO0 GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO15

USB_DP USB_DM SWDIO SWCLK RUN BOOTSEL

GPIO29 GPIO28 GPIO27 GPIO26 GPIO25 GPIO24 GPIO23 GPIO22 GPIO21 GPIO20 GPIO19 GPIO18 GPIO17 GPIO16

RP2350B Core2350B

RIOW RIOR AD0 AD1 AD2 AD3 AD4 AD5 AD6 AD7 RA8 RA9

UART_TX ADS RI/OCHRDY RV_CLK RV_DATA RDRQ RIRQ RTC RDACK LRCK BCK DIN

470R LED GND

SWD BOOTSEL

SAMPLE RAM: on the module

The Core2350B2 has 8 MB of PSRAM on the RP2350B's QMI (CS1 = GPIO47, memory mapped). It replaces v2.0's U8 (APS6404L on GPIO0–3, PIO SPI); GPIO0–3 are now free.

IRQ/DMA SETTING

IRQ/DMA

IRQ2 IRQ3 IRQ4 IRQ5 IRQ7 DRQ1 DACK1 DRQ3 DACK3

IRQ DRQ DACK

I2S DAC

Audio

BCK DIN LRCK

BCK AUDIO_L DIN AUDIO_R LRCK

PGAUDIO_L PGAUDIO_R

File: audio.kicad_sch

POWER

USB-A host

PWR_FLAG +5V

VBUS D+ D- Shield GND

0.5A PTC

3_USB_D+ 2_USB_D-

+5V 1N5819 D3 PWR_FLAG

VBUS_PRG 1N5819

MCP1826S-3302E/AB VSYS VI VO GND

10uF 10uF GND GND GND

AVDD 100uF 0.1uF AGND

WTAUDIO_L WTAUDIO_R RV_DATA RV_CLK

2.2uF film 2.2uF film 2.2uF film 2.2uF film

AVDD 1ch_IN 2ch_IN Vcc 1ch_OUT 2ch_OUT DATA CLK GND M62429L AGND

Mixer

WTVOINL_LD WTVOINL_RD

File: mixer.kicad_sch

SJ1-3535NG-GR TIP RING SLEEVE AGND

Ferrite bead Ferrite bead AVDD PWR_FLAG GND AGND PWR_FLAG

Waveblaster

UART_TX +5V +12V WTAUDIO_R -12V WTAUDIO_L WT_RESET GND AGND

ADDR/DATA MUX/DEMUX

SN74CB3T3257

S 1B1 1B2 1A 2B1 2B2 2A GND VCC OE 4B1 4B2 4A 3B1 3B2 3A

ADS A0 D0 AD0 A1 D1 AD1 GND

BUSOE A2 D2 AD2 A3 D3 AD3

SN74CB3T3257

ADS A4 D4 AD4 A5 D5 AD5 GND

BUSOE A6 D6 AD6 A7 D7 AD7

MISC LEVEL SHIFT/BUFFER

74AHC244

TC GND A8 A9 RDRQ RIRQ GND GND BUSOE

1A0 1A1 1A2 1A3 2A0 2A1 2A2 2A3 1OE 2OE VCC GND

1Y0 1Y1 1Y2 1Y3 2Y0 2Y1 2Y2 2Y3

RTC RA8 RA9 DRQ IRQ

AEN IO MASKING

Disables IOR and IOW signals when other DMA in system happens

74AHC14 74AHC00

IOR AEN DACK IOW +5V

RIOR RIOW

RESET CIRCUIT

Latches BUSOE high after ADS is toggled for the first time, sets low on RESET

74AHC14 74AHC00

RESET 10k ADS 1k 10k BUSOE RUN 0.1uF GND RUN WT_RESET

I/OCHRDY OPEN DRAIN

U6B 74AHCT126 I/OCHRDY RI/OCHRDY GND

MOUNTING

MountingHole MountingHole GND

DECOUPLING CAPS

+5V 47uF 0.1uF 0.1uF 0.1uF 0.1uF 0.1uF 0.1uF 1uF 10uF GND GND

74AHCxx LOGIC PWR

U6E 74AHC14 74AHC00 74AHC00 74AHCT126 +5V C46 0.1uF GND GND

DACK INVERT

Saves PIO instructions due to JMP limitations

74AHC14 DACK RDACK

PWR_FLAG +12V PWR_FLAG -12V

U6C 74AHCT126 GND GND

U6D 74AHCT126 GND GND

U6 74AHCT126 on +5V: OE high pulls the output low, OE low leaves it open (replaces the 74LVC2G06). Spare gates: A and OE tied low.

MIDI/DEBUG OUT

UART_TX 74AHC00 GND UART_TX

+5V 220R Ferrite bead

U6A 74AHCT126 GND 220R Ferrite bead

T R S SJ1-3535NG GND

U12B

Core2350B

3V3 3V3_EN VBUS ADC_VREF Core2350B power, GPIO30-47

GPIO30 GPIO31 GPIO32 GPIO33 GPIO34 GPIO35 GPIO36 GPIO37 GPIO38 GPIO39 GPIO40 GPIO41 GPIO42 GPIO43 GPIO44 GPIO45 GPIO46 GPIO47

GND GND GND GND GND GND GND GND

VSYS

U12 module power: VSYS feeds the module's VBUS pin and its own LDO (ME6217, 3V3_EN pulled up on the module) powers the module. The module 3V3 pin is not connected: U9 powers the card logic, DAC and PSRAM. GPIO30–47 are not used (GPIO39: module LED, GPIO47: module PSRAM CS on Core2350B1/B2).

USB-C PROGRAM PORT

USB-C program

J6

VBUS CC1 CC2 D- D- D+ D+ SHIELD GND SH

A4 B5 A7 B7 A6 B6 A8 B8 A1

VBUS_PRG R19 5.1k USB_D- USB_D+ SBU1 SBU2 GND

5.1k R20 GND

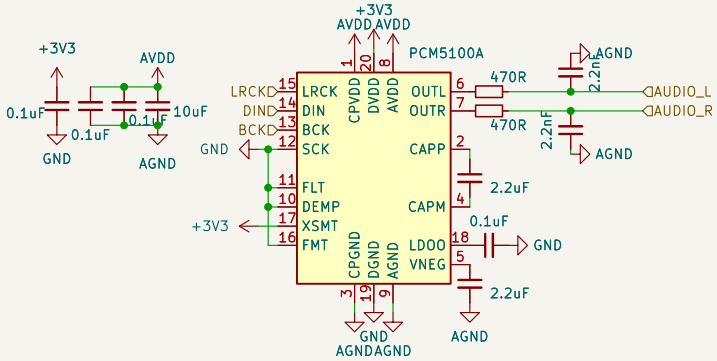
5.1k Rd on CC1/CC2: device only, never sources VBUS. D+/D− shared with J5 (USB host) as on v2.0.

v2.0 ISA logic and GPIO map; TSSOP/SOIC-only chips on DIP adapters
Through-hole PicoGUS 2.0 on a Waveshare Core2350B (RP2350B) module

Sheet: /
File: PicoGUS-THT.kicad_sch

Title: PicoGUS v2.1 THT

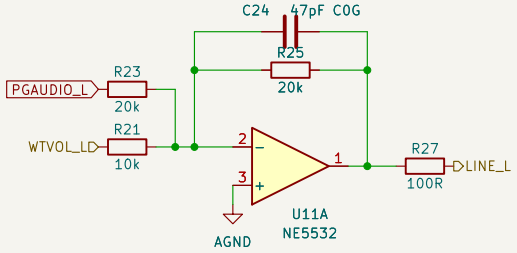
Size: A3	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 1/3

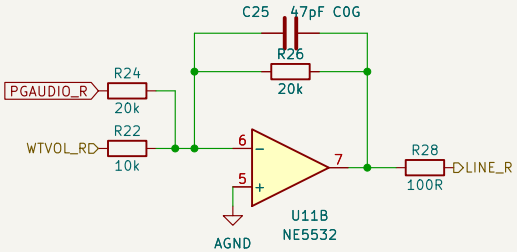


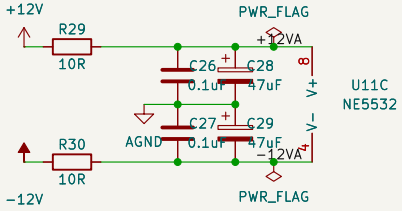
Sheet: /Audio/		
File: audio.kicad_sch		
Title: PicoGUS v2.1 THT – PCM5100A DAC		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 2/3

LINE-OUT MIXER

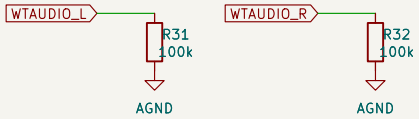
Inverting summing amplifier from THT v1.3: Waveblaster (after U13 M62429 volume) x2 = +6 dB through R21/R22 10k.
PCM5100A x1 through R23/R24 20k, feedback 20k || 47 pF (pole ~170 kHz). U11 runs from ISA +/-12V through 10R / 47uF.
The DAC output is ground-centred, so the PCM path is DC coupled (no electrolytic); the M62429 output sits at about
VCC/2, so C41/C42 (film) couple it. Both channels invert, so left and right keep their relative phase.







WT input pull-downs (THT v1.3 R14/R15)



Sheet: /Mixer/		
File: mixer.kicad_sch		
Title: PicoGUS v2.1 THT – line-out mixer		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 3/3